

Code-Layout, Readability and Reusability

Date	08 October 2026
Team ID	SWTID-2026-2386
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4 spaces in Python, 2 in HTML/JS/CSS
2	Proper File Structure	Files and folders are logically organized	Yes	main.py routes, gemini_utils.py Gemini service, one module per feature, config.py, models.py, templates/, static/, tests/, tools/
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. answer_index, lastAnswer, models_called
4	Function / Method Names	Functions are descriptively named	Yes	e.g. answer_question, explain_concept, generate_quiz, clean_json_block, get_learning_recommendations
5	Code Comments	Inline and block comments explain logic	Yes	Module docstrings and comments for non-obvious logic
6	Modular Design	Code is split into reusable functions/modules	Yes	One module per feature and one shared Gemini service
7	No Redundant Code	Duplicate or unused code is removed	Yes	All five features share gemini_utils.generate; all five endpoints share _validated_text and _run
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Empty/long input (400), invalid request (422), missing or rejected key, Gemini failure, rate limit (429), bad quiz output (502)

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	gemini_utils.generate() with model chain	Python	All five feature modules	High
2	_validated_text() and _run()	Python / FastAPI	All five endpoints	High
3	GeminiError	Python	Service, modules, API error handler	High
4	UserRequest model	Python / Pydantic	All five endpoints	High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
5	<code>clean_json_block()</code>	Python	Quiz module (reusable for any JSON reply)	Medium
6	<code>showError()</code> / <code>showText()</code> in <code>app.js</code>	JavaScript	All five tasks	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation of routes, service, modules, models and configuration
Readability	4	Short functions and descriptive names; <code>quiz_module.py</code> holds the most logic
Reusability	5	One Gemini call shared by every feature
Documentation / Comments	4	Docstrings, README and phase documents
Overall Score	4.5	Self-assessment by the team